Rohan Malhotra

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EXPERIENCE

IBM Dallas, TX

Applications Developer Intern, IBM Consulting *May 2026 – Aug 2026*

- Built an XGBoost cash-flow model and 13-day rolling cash forecasts in Oracle EPM Planning for finance teams.
- Built SOFR loan, interest-accrual, and close-report models; piped banking API data into ERP and EPM as JSON.
- Fine-tuned Qwen-Coder-32B on Oracle EPM assets, raising EPM build-generation accuracy from **36.7% to 95.0%**.

Robotics Engineer Intern *Jun 2026 – Aug 2026*

- Labeled 898 images and trained YOLO11 models to 99.5% mAP@50 with 99.4% precision and 99.4% recall.

Kalshi New York, NY

Quantitative Developer *Jan 2026 – May 2026*

- Modeled job-loss risk as a Poisson hazard on FRED and BLS data; measured hedge tail-risk cuts via Monte Carlo.
- Built a Next.js/Python engine that recommends hedge size and compares cost vs. tail risk using FRED and BLS data.

Aress Software New York, NY

Machine Learning Intern *Jun 2025 – Aug 2025*

- Built Power BI dashboards surfacing 20+ real-time KPIs for global operations teams; managed the JIRA workflow.
- Designed NoSQL pipelines unifying 10+ data sources into one analytics layer; optimized SQL queries for speed.
- Trained and deployed PyTorch models on **500K+ multimodal records** across Azure and AWS.

Y-Axis Overseas Careers Chatham, NJ

Data Engineering Intern *May 2024 – Aug 2024*

- Cleaned and standardized 20K+ SQL records with schema fixes and validation, cutting data-prep time by **15%**.
- Engineered 12 predictive features from dataset analysis and transformations, raising model accuracy by **10%**.
- Built automated Excel dashboards with dynamic filtering, enabling faster eligibility reviews.

RESEARCH

Reddit Data in Quantitative Financial Models (published) Jan 2025 – May 2025

Co-author, Virginia Tech (VTechWorks) *hdl.handle.net/10919/124730*

- Applied NLP sentiment scoring and regression models in Python to quantify sentiment's impact on equity returns.

An Economic Approach to Optimize Capital Allocation (published) Aug 2024 – Dec 2024

Author, Virginia Tech (VTechWorks) *hdl.handle.net/10919/124871*

- Ran Monte Carlo tests and backtests in Python comparing full vs. fractional Kelly under drawdowns and costs.

PROJECTS

Love Island Sponsorship ROI Analysis | *Python, Pandas, Plotly* 2025

- Compared sponsorship cost, reach, and engagement across Love Island brand deals; built an interactive ROI page.

SPY Gradient Boosting Predictor | *scikit-learn, PCA, yfinance* 2025

- Trained a gradient-boosting model to predict the next SPY close from EMA, RSI, and MACD features with PCA.

EDUCATION

New York University, Courant Institute of Mathematical Sciences New York, NY

B.A. in Mathematics and Computer Science (Double Major) *Expected May 2027*

- **GPA:** 3.7/4.0 | Accelerated three-year degree | **Graduate course:** Mathematical Techniques for CS Applications
- **Coursework:** Probability & Statistics, Machine Learning, Linear Algebra, Numerical Methods, Algorithms

SKILLS

Languages: Python, SQL, R, C++, MATLAB, Java

Data & ML: Pandas, NumPy, SciPy, statsmodels, scikit-learn, XGBoost, PyTorch, NLP, regression, Monte Carlo

Tools: Power BI, Excel (macros), PostgreSQL, NoSQL, Jupyter, Plotly, AWS, Azure, Docker, Git, Oracle EPM